# Annihilation Rates of Methl Halides Explained with Vibrational Feshbach Resonance

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$Z_{\rm eff} \gg Z$ ($Z_{eff} \gg Z$).

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$Z$

When a positron and an electron collide, their masses ~~will~~ can be converted into energy in the form of photons in a process known as annihilation [1]. The annihilation rate of slow positrons interacting with molecules was initially expected to scale linearly with molecular size, and hence the number of electrons, Z. However, since the 1960s, experiments have shown that it increases exponentially[2], as if the molecule effectively has many more electrons, ~~denoted~~ namely $Z_{\text{eff}}$. Various explanations were proposed over the following decades — including the formation of virtual positronium [3], weakly bound positron states, and resonances[4] — but a unifying understanding remained elusive. It took half a century for theory and experiment to align, a milestone achieved by ~~Gleb~~ Gribakin and ~~C.M.R.~~ Lee with the publication of "Positron Annihilation in Molecules by Capture into Vibrational Feshbach Resonances of Infrared-Active Modes[5]." full stop after quotes.

The paper examines experimental annihilation rates for the methyl halides $CH_3F$, $CH_3Cl$, and $CH_3Br$[6], applying a theoretical framework based on Breit-Wigner formalism first proposed ~~by Gribakin~~ years earlier[7]. Two mechanisms of annihilation are considered: direct and resonant. In direct annihilation, the positron is not bound to the molecule and annihilates while still "in flight." In resonant annihilation, the positron's energy is absorbed by a vibrational excitation of the molecule, capturing it in a positron-molecule ~~complex~~ state called a vibrational Feshbach resonance (VFR), after which it will either detach or annihilate. This leads to sharp peaks in $Z_{\text{eff}}$ which correspond to the vibrational resonances of the molecule. the energy dependence of

If positrons are indeed captured in resonances, peaks in $Z_{\text{eff}}$ should be downshifted by the binding energy compared to the ~~positron~~ vibrational energy spectrum, as to be captured in the resonance, the positrons need to ~~use~~ release energy to bind to the molecule. Moreover, the greater the binding energy, $\epsilon_0$, the harder it is for the positron to detach, increasing $Z_{\text{eff}}$. No binding energy was known for any of the se molecules, so it was chosen as a free parameter of the theory. Adjustment of $\epsilon_0$ aligned the theoretical curves with the experimental data, with $\epsilon_0 = 0.3$meV, 25meV and 40meV for $CH_3F$, $CH_3Cl$, and $CH_3Br$ respectively. Due to the complex pattern of the peaks, the fact that simply changing this single parameter creates ~~great~~ good agreement with experiment is ~~good~~ strong evidence of the theory's validity.

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This agreement is, in part, due to the molecules chosen. They only have six vibrational modes, all of which are infrared active[8]. Infrared active modes have changing dipole moments, which provides ~~increases~~ dipole coupling between the molecule and the positron, making it easier for the positron to bind. Additionally, ~~they~~ molecules are big enough to have bound states that allow VFR, but small enough that an excited state will not relax into multiple vibrational modes [9]. A model incorporating the latter was the next step for the theory, which was successfully developed in 2017[10]. The current ~~focus is on~~ challenge is to develop a theory that does not rely on a long-range dipole coupling approximation, with molecules exhibiting peaks at infrared inactive vibrational modes [11] (and hence, non-dipolar). ~~casting doubt on its applicability~~.

better use $\varepsilon_b$ ($\varepsilon_b$)

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